Cheat sheet · Chia U-Meng Adrian (Xie Youming) · MTCS-MRMC5

Three messages

1. Per-client rate limits cannot see a low-and-slow attack that rotates IPs; one-minute campaign-level aggregates can.
2. A 28,241-parameter TCN learns the campaign rule and carries it to unseen traffic, even when the IP pattern flips.
3. It costs about 0.65 ms once a minute (≈ 0.001% of one CPU core).

Key numbers

Item	Values
Training Dataset	53.1 h · 8,095,202 requests · 3,187 windows · split 70/15/15 in time order
Chaotic Dataset	154 min · 145 sequences (44 attack / 101 benign)
Large Dataset	233.4 h · 33,731,232 requests · 13,351 sequences (1,992 / 11,359)
Large Dataset, θ = 0.50	precision 0.9945 · recall 1.0000 · F1 0.9972 · FPR 0.00097 · 1.39 alerts/day · ROC-AUC 0.999983
Chaotic Dataset, θ = 0.50	precision 0.8947 · recall 0.7727 · F1 0.8293 · FPR 0.0396 · ROC-AUC 0.9876
Hold-out test set	470 benign sequences · 0 false positives · max score 0.000000146
Plot twist (mean IPs)	Training 1,459 attack vs 775 benign · Large 288 vs 803 · Spearman: failure rate +0.707, IPs −0.392
Earlier model	497 windows, 0.40 labels, GELU · Large Dataset recall 0.2575, ROC-AUC 0.3576
Latency	mean 0.65 ms · P50 0.62 · P95 0.81 · P99 1.64 · ≈ 0.001% of one core
Errors	15 FPs (failure rate 0.19–0.30, all ≤ 15 min from an attack) · 10 misses (0.32–0.36, last Chaotic attack)
Model	3 residual blocks, dilations 1/2/4, k = 3, channels 16/32/64, dropout 0.2, ReLU · 28,241 parameters · RF 29
Training	BCE with w+ = 0.3925 · AdamW lr 0.001 · batch 64 · 30 epochs · epoch 27 kept · no seed
Label rule	failure rate > 0.30 and distinct IPs > 5 (last window of the sequence)

Limitations (ch.5.2)

L1 labels are a rule over the inputs · L2 validation set is benign only · L3 evasion: ≈ 2.33 good logins per failure · L4 overlapping sequences, single run, no seed · L5 JA3 constant, no header features · L6 synthetic traffic · L7 minutes, not IPs; no throttle built

Answer pattern

Acknowledge the point in one sentence. → Give the evidence: number, table or chapter. → State the boundary: what the result does and does not show. → Offer the backup slide if useful.